SMART GUARDIAN

PROJECT NAME:	Smart Guardian
PROJECT GROUP:	Vedika Raut

WHY THIS PROJECT ?

- The Smart Guardian project is developed to enhance safety and security by automatically detecting threats like human intrusion and fire, reducing the need for continuous human monitoring.
- The main objective is to enhance security and protection by continuously monitoring critical parameters and responding instantly to abnormal conditions.

INTRODUCTION

- Smart Guardian is an IoT-based intelligent surveillance and safety system that monitors surroundings in real time using camera and sensors and provides instant alerts during emergencies.
- The system continuously senses, analyzes, and transmits data to ensure timely alerts and preventive action.

KEY FEATURES

- Real-time video streaming
- Fire detection using smoke and temperature sensors
- Instant alert notifications
- YOLO-based human detection
- Allows easy replacement and upgrading of components.
- Low-cost and scalable system

ARCHITECTURE

- The system uses an ESP32-CAM to capture and stream live video.
- The video stream is processed using Python-based YOLO for real-time human detection.
- Fire detection is achieved using MQ smoke sensor and DHT11 temperature sensor to monitor smoke concentration and temperature rise.
- Alerts are sent to the user through internet-based notification services.

RESULTS

- The Smart Guardian system successfully detects human presence and fire conditions in real time and sends timely alerts, ensuring improved safety and surveillance.

FUTURE SCOPE

1. **Cloud-based data storage , AI-based threat prediction**
2. **Deployment in smart homes, industries, and drones**
3. **Integration of face recognition**